Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief

Access and independence

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): this planning work is [QUAL] Biology core-practical ground; the same enzyme and method skills are used in food and drink production and in hospital and diagnostic laboratories, and lead on to Level 3 science and laboratory apprenticeships.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Use paper models or teacher-authorised apparatus. No tasting, personal genetic disclosure, physical contact or unplanned chemical handling. Offer adult-directed handling and a seated route.

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TA brief · continued

Prompt ladder

Prompt ladder for adults

1. Point to the relevant place / instruction.
2. Ask one short question: "What is it asking you to find?"
3. Rephrase the instruction in simpler language.
4. Provide a scaffold or limited choice.
5. Model the first step only, then return the task to the pupil.

Planned adaptations

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own [SYS] target on the school's evidence app; the [SYS] carries their EHCP-linked and [SYS] targets, so one record serves every plan.

Follow the pupil's [SYS] where it differs from these routines.

00–04 · Arrival task

At arrival, say the spoken line on the marking card once. Use the marking card's spoken arrival guidance once. Read the response with the help used in view. Make one same-lesson adjustment by about minute eight. IF NEW — 1 min read local facts; 2 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min sample and reveal.

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TA brief · continued

04–08 · Starter · what can iodine tell us?

No reagents needed today. Read labels aloud: colour is not the only clue. Choose the arrival adjustment from the first-lesson-back card. IF NEW — 2 min show and name outcomes; 1 min compare; 1 min check endpoint. IF REVISION — 1 min inspect; 2 min justify outcomes; 1 min resolve the endpoint.

08–13 · I do · bind, break down, release

Pause each step. This is a simplified shape model, not literal rigid molecules. Enzymes can flex; amylase is not a living organism. IF NEW — 3 min model all three moments; 1 min name terms; 1 min pupil retells. IF REVISION — 1 min orient; 3 min predict each step; 1 min repair an explanation.

13–18 · We do · label the enzyme model

Named involvement: invite a pupil's label choice and engage with the explanation. Select-card then destination works with keyboard/touch; paper labels give the same science. IF NEW — 1 min read parts; 3 min match and explain; 1 min check. IF REVISION — 0.5 min orient; 3 min label and justify; 1.5 min compare.

18–22 · I do · connect pH to the measure

Do not assume pH 7 is a universal optimum. Reduced activity alone does not prove permanent denaturation. A shorter endpoint time implies higher relative rate only for matched amounts/endpoints. IF NEW — 2 min connect pH and active site; 1 min link to time; 1 min check. IF REVISION — 1 min orient; 2 min explain causal chain; 1 min distinguish claim and evidence.

22–26 · We do · organise the investigation

F/B may swap if both precede mixing. This is planning, not permission to handle reagents. The later lesson needs the centre method and risk assessment. IF NEW — 1 min read six cards; 2 min organise; 1 min explain a dependency. IF REVISION — 0.5 min read; 2.5 min organise and justify; 1 min check.

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TA brief · continued

26–34 · You do

Use supplied headings and card letters rather than rewriting a long method. Switch routes at the task with no record. Pointing, annotation or pupil-directed scribing preserve the scientific demand. IF NEW — 2 min use reference; 6 min complete five concise points; 2 min use one precise feedback point. IF REVISION — 1 min use reference; 7 min complete five concise points; 2 min use one precise feedback point.

34–37 · Review and improve

Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.

37–40 · Exit ticket

Use a book response OR one ticket. The pupil owns the response; no counter-initial or extra receipt. Use the evidence to choose last lesson's one next teaching move. IF NEW — 1 min read evidence; 2 min respond; 1 min engage against model answer. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min engage against model answer.

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TA brief · continued

Planning and evidence

Say once: "I haven't seen what you've been doing… I start from what you show me".

Lesson code: SCI-LAUNCH-A1-W08-L1. SoW reference: LAUNCH Science, Autumn 1 return block; L1.

Exact SoW outcome: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Evidence (scheme of work): [AWARD] [QUAL] Paper 1 ([QUAL]) topic tests; [AWARD] [QUAL] science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Use enzyme, substrate, active site and specificity accurately in an explanation.; Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity.; Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables.

Evidence location: the pupil's explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L1. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Staff: confirm the planned adaptations actually used and record any changes below. [SYS] strategies are linked to actual tasks above.

60-second recap of the previous lesson: Topic 1 Exam Practice. Topic 1 review & exam-style questions. Pause and recall one idea you already know; no particular old question is required. In an exam-style response, follow the instruction, use accurate subject terms and connect a reason to the given evidence. If an idea is unclear, use today's clues and ask for one small prompt.

Must teach core: I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Use enzyme, substrate, active site and specificity accurately in an explanation.

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TA brief · continued

Feedback and evidence

No cold calling; nobody is asked to read difficult text aloud.

Staff mark I, S or C beside this task.